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Module: 7PAM2016 Advanced Research Topics in Data Science

GitHub Link: [Please click this link to see code on Github:-\)](#)

Google Colab Link: [Please click this link to see code on colab:-\)](#)

1. Introduction

To generate synthetic data by learning underlying data distributions Generative Adversarial Networks (GANs) are powerful deep learning models. In this report, different applications of GANs across multiple domains being investigated and implemented including synthetic 2D data, medical imaging, cybersecurity traffic, and sketch generation.

The main objective is to evaluate and test the capability of GANs that how GANs perform under varying levels of complexity and to analyse their strengths and limitations. Both methods qualitative in the form of visual outputs and quantitative in the form of evaluation metrics such as FID used to assess performance.

2. Methodology Overview

Two neural networks of a Generative Adversarial Network (GAN) trained simultaneously in a competitive setting. To resembles real samples, as input the generator takes random noise and attempts to produce synthetic data while the main function of the discriminator is to evaluate whether a given sample is real or generated. During the training process the discriminator learns to better distinguish real data from fake data and differentiate while the generator improves by trying to fool the discriminator. This process which is adverse continue until the generator successfully produces the images that are increasingly realistic effectively learning the underlying data distribution.

a. Architecture Justification

Based on the data types different architectures were selected which are as follow:

- **2D Data:** Due to low dimensionality fully, connected networks were used.
- **Images:** To capture spatial features DCGAN called as convolutional layers were used.
- **Tabular Data:** For feature vectors fully, connected layers were used.

This ensures that the model structure aligns with the nature of the data.

3. GAN on Synthetic Data

3.1. Sine Wave Distribution

The GAN was trained to learn a sine wave distribution as seen in the **Figure 3.1**.

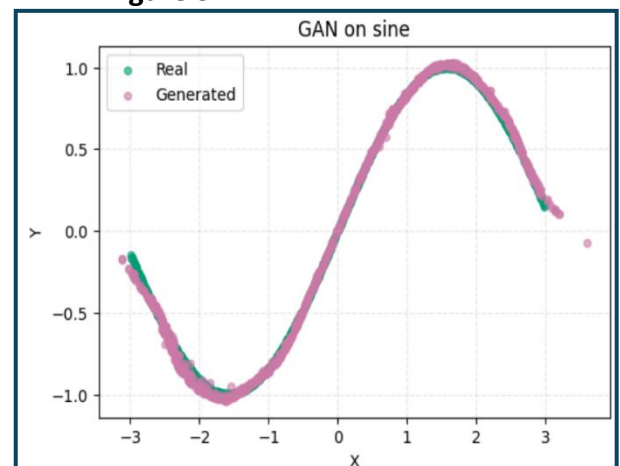


Figure 3.1 Sine Wave Distribution

The comparison between the real and Generated samples of the sine wave distribution is shown in the

Figure 3.1. As shown in the figure the Green is the Real Sine wave while the Red is the generated Sine Wave which is indicating that the underlying distribution successfully learned by the GAN as the generated data closely follows the shape of the true sine curve. Due to limited data coverage and model approximation minor deviations are observed at the boundaries. Overall, in capturing the continuous structure of the sine function the generator demonstrates strong performance.

3.2. Gaussian Mixture Distribution

The GAN was applied to a mixture of Gaussian as seen in the **Figure 3.2(a)** & **Figure 3.2(b)**.

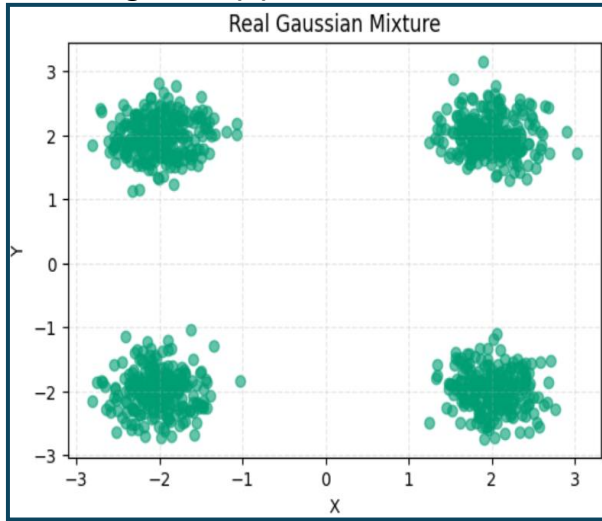


Figure 3.2(a) Real Gaussian Distribution

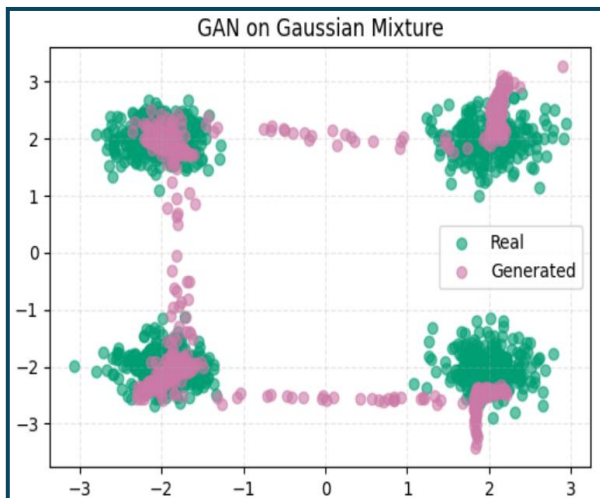


Figure 3.2(b) Real vs Generated Gaussian Distribution

As seen in the **Figure 3.2(a)** the four Gaussian clusters located in separate regions of the 2D space consists of real data. The **Figure 3.2(b)** shows that the GAN learned the multi-modal structure of the dataset successfully as the generated samples reproduce the main cluster locations. However, the presence of several generated points appear between clusters indicating incomplete modelling of cluster boundaries and minor instability.

3.3. Modified GAN Architecture

To Improve the Learning capacity a deeper network architecture was implemented as seen in the **Figure 3.3 (a)** & **Figure 3.3 (b)**.

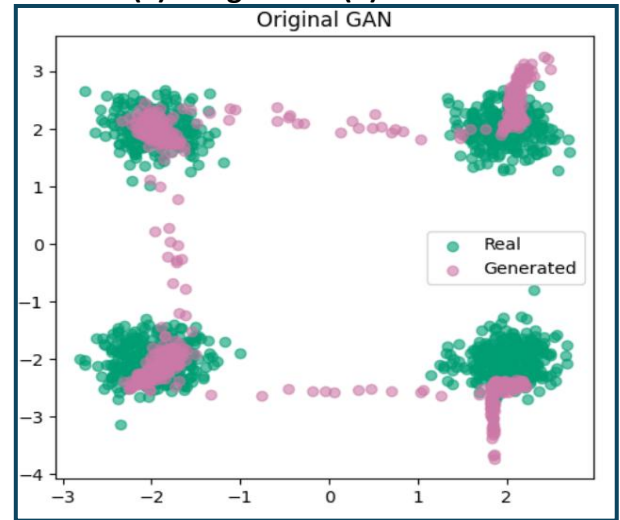


Figure 3.3 (a) Real GAN

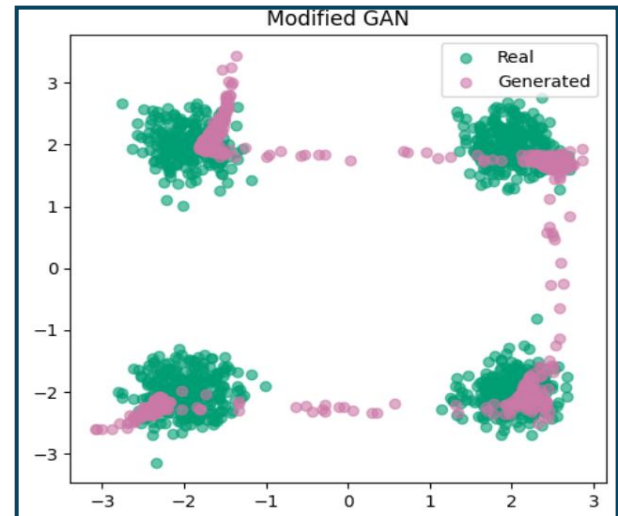


Figure 3.3 (b) Modified GAN

As seen in the above figures the modified GAN produced slightly more compact generated clusters and maintained the ability to learn the four Gaussian modes while compared to the original GAN. However, while stability improved marginally, some interpolated samples between clusters were still observed indicating that the model still not fully learned the boundaries, and the overall performance remained the same to the original model. The **FID score** for Real and modified GAN is as follow:

- Original GAN FID Score: 0.092
- Modified GAN FID Score: 0.03

4. BloodMNIST (Medical Imaging)

4.1. Dataset Exploration

The dataset called BloodMNIST consists of gray scale images of blood cells across multiple classes.

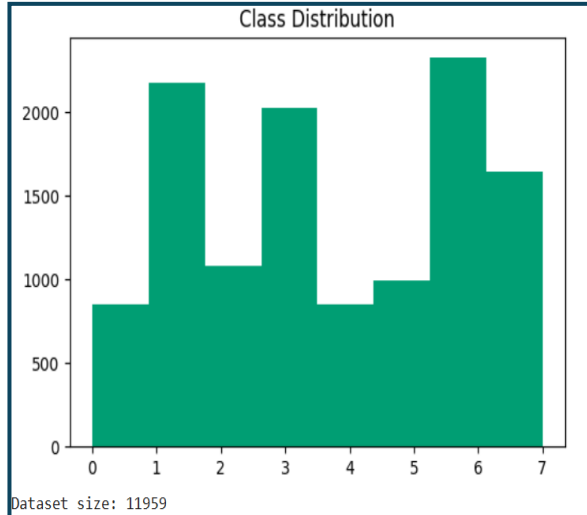


Figure 4.1(a) Class Distribution

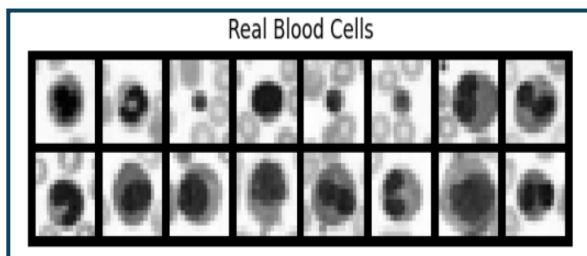


Figure 4.1(b) Real Blood

4.2. Model and Training

To capture spatial features a DCGAN architecture was used with convolutional layers. To observe convergence behavior training losses were monitored.

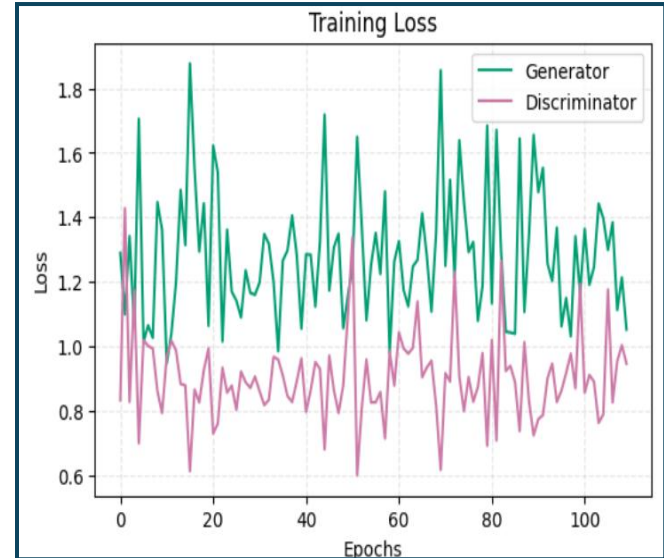


Figure 4.2. Training Loss Curves

This plot shows the losses changes over 110 epochs in the GAN. In the GAN training the generator tries to fool the discriminator while the discriminator tries to be stable in the training and distinguish between real and generated samples, that's why both curves fluctuate, which is normal because GAN training is inherently unstable. The loss trend of the generator generally varies without a clear downward appearance, meaning it's improving at producing realistic samples, while the discriminators adapts to the generator's progress to generate fake samples rises and falls as shown in the figure. Overall, the graph reflects a adversarial training dynamic which is typical where neither model converges smoothly but instead competes throughout training.

4.3. Results & Comparison

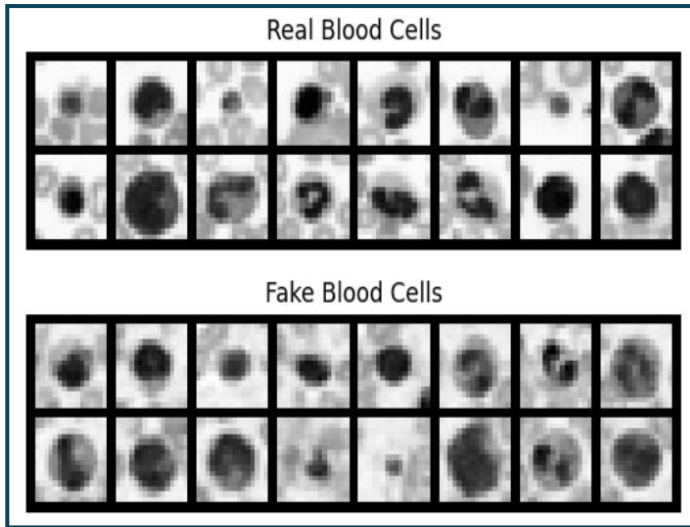


Figure 4.3 Real Vs Fake Cells

The **Figure 4.3** shows the comparison between Real and Fake Blood Cells after the training of the model. The overall data distribution represented by the key features such as circular shapes, dark centres, and general cell structures learned by the model successfully. However, the generated images are blurrier and less detailed than the real samples, with some distorted shapes and the quality of the generated images is not as much clear as compared to the real images. Overall, the GAN still lacked fine visual quality but produced realistic-looking samples.

4.4. Evaluation (FID) & Discussion

The model achieved a score of **FID 112.7** which shows moderate similarity between the real blood samples and fake blood samples. The model struggles with fine details but partially captures structural patterns. This is due to low resolution of images and low or limited model capacity.

5. CICIDS 2017 (Cybersecurity)

5.1. Dataset Exploration

A strong class imbalance was observed while the dataset consists of network traffic data with multiple attack types.

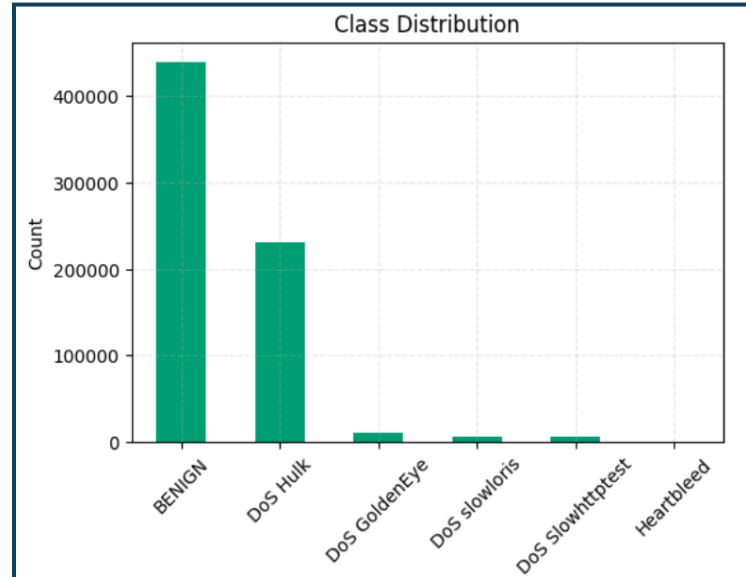


Figure 5.1 Class Distribution

The graph represents the cybersecurity dataset and the class distribution of the CICIDS 2017 cybersecurity dataset used in this project. The attack classes such as DoS GoldenEye, DoS slowloris, DoS Slowhttptest, and Heartbleed have significantly fewer samples while the BENIGN traffic class contains the highest number of samples, followed by DoS Hulk. Normal traffic dominates and some attack types are underrepresented this shows a strong class imbalance in the dataset. The GAN may learn the majority patterns from the major attacks more effectively than minority attack behaviours and such imbalance can make model training more difficult.

5.2. Model & Training

As seen in the **Figure 5.2** using an adversarial GAN framework a model

was trained in which the generator learns to produce synthetic samples while the discriminator learns to distinguish real data from generated data and the discriminator loss decreases steadily and stabilises at a low value during the training which indicated that it quickly became effective at classification. In opposite to this, the gradual loss increase of the generator over epochs suggests that the generator found it increasingly difficult to fool the discriminator. The pattern of the training shows that an imbalance where the discriminator became stronger than the generator which is resulting in limited generator improvement.

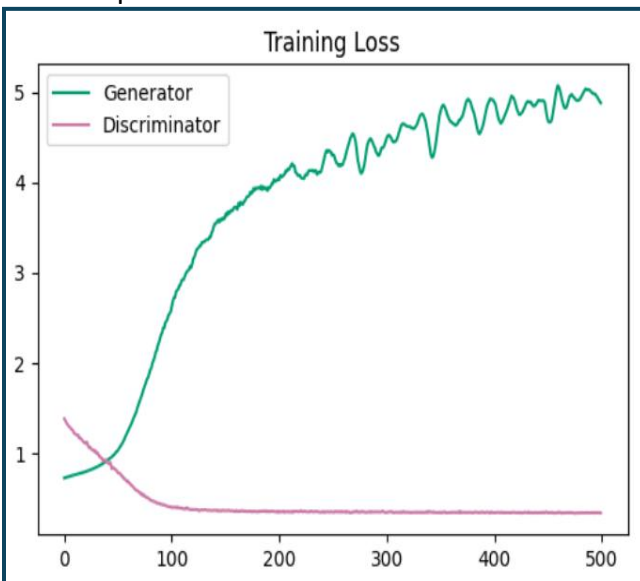


Figure 5.2 Training Loss Curves

5.3. Results & Discussion

To compare real and generated data dimensionality reduction techniques were used. Such as T-SNE real vs fake traffic plot shows the comparison between both traffics.

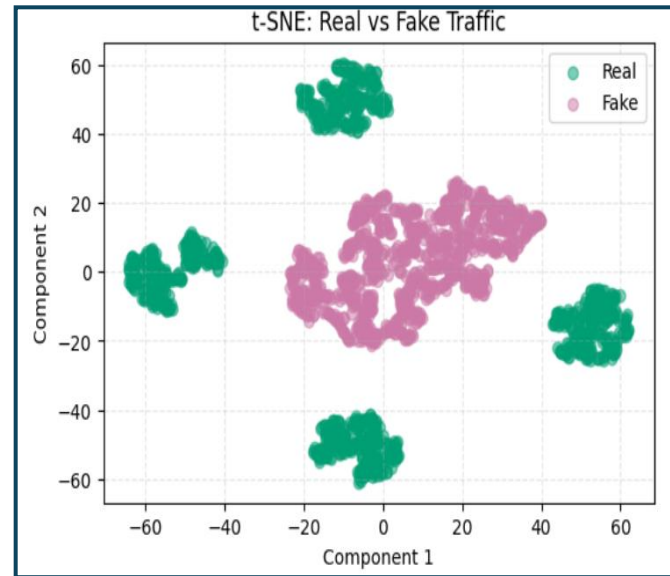


Figure 5.3 t-SNE Real vs Fake traffic

The graph shows comparison between real traffic in green colour and the GAN generated traffic in the pink colour. T-SNE allowing patterns to be visually compared while reducing the high-dimensional feature space into two dimensions. The real data reflecting the diversity of traffic behaviours in the dataset as it appears in several separated clusters. The generated clusters remain clearly distinguishable from the real clusters but are concentrated in one main region with partial overlap. This shows that the GAN did not fully capture the complete complexity and multi-modal structure of the real distribution but learned some general characteristics of the traffic data. The **FID score of 96.56** further support this interpretation, which indicates although a noticeable distribution gap still remains but some improvement in similarity between real and generated samples.

The GAN struggled to learn the full distribution of the data due to:

- Class imbalance
- High dimensionality
- Complex attack patterns

6. QuickDraw

6.1. Pizza Dataset

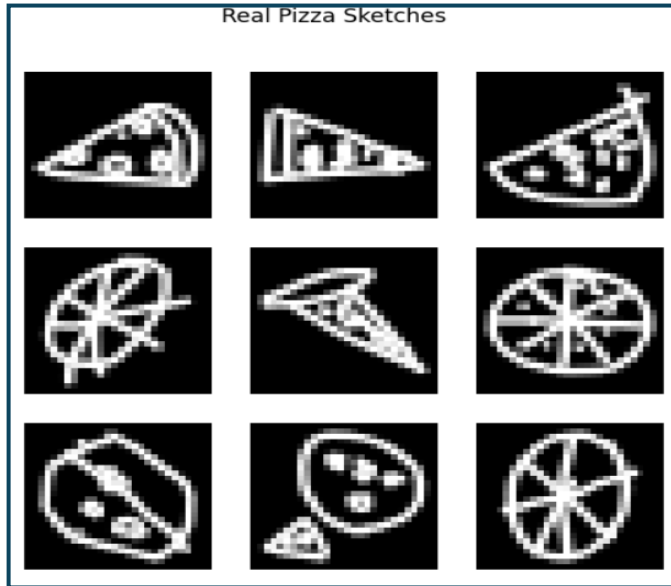


Figure 6.1 Real Pizza Dataset sketches

6.2. Model Training



Figure 6.2 Training loss

The above figure shows the generator loss remains high and fluctuates between approximately 3.5 and 7, and unstable GAN training behavior while the loss of discriminator stayed almost at 0 throughout the training. It shows the

discriminator is making difficult for the generator to improve as it became too strong very quickly and easily distinguished fake samples. The sudden drop and recovery of the generator loss suggests temporary instability during training.

6.3. Results & Discussion

The **Figure 6.3** shows the comparison between real and fake(generated) pizza sketches.

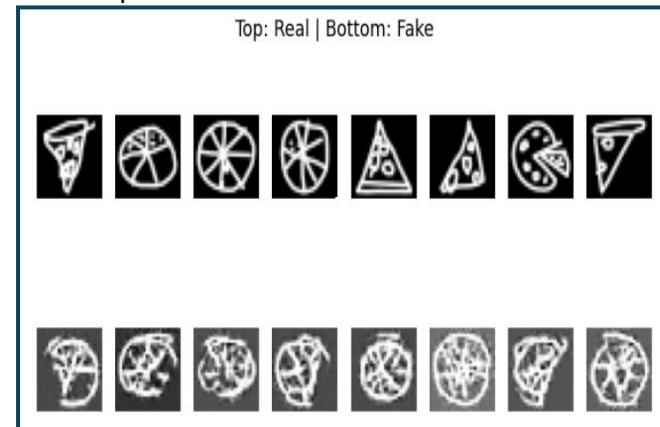


Figure 6.3 Comparison of Real vs Fake

The **Figure 6.3** represents the comparison of real Pizza sketches (top) and the GAN generated pizza sketches(bottom). It is showing that the model learned the main structural features of the dataset as the synthetic images successfully capture the overall circular pizza shape and internal slice-like patterns. However, with some distorted outlines the generated sketches are blurrier and less detailed than the real samples. Overall, the GAN did not fully match the clarity and precision of the original sketches but produced recognisable pizza-like images. For evaluation purposes a **FID** test conducted and the model achieved a score of **115.60** which shows the moderate similarity between the real and generated images with room for further

improvement. **The extension of Part 2.3 has been done using Cup dataset and given a FID score of 64.18.**

7. Evaluation Metrics

The method used to evaluate the results and the similarity between real and generated images is the Fréchet Inception Distance (FID). As the lower values indicate better performance but the values obtained shows the moderate performance where GAN captures the general pattern but struggles with fine details.

Below is the FID comparison Table of all parts:

Sr.	Dataset Name	FID Score
1.	Original GAN & Modified GAN	0.092 & 0.03
2.	BloodMNIST	112.71
3.	CICIDS	96.56
4.	Pizza	115.60
5.	Cup (Extension Part 2.3)	64.18

8. Conclusion

The study shows that the GAN performance declines or become weak when applied to more complex, high-dimensional, or highly variable datasets but are effective in modelling simple and structured data distributions. Factors such as network architecture, dataset complexity, training duration, and the overall stability of the adversarial learning process strongly influenced the quality of the generated outputs.

Future works could focus and open new ways to improve generation quality through different factors such as use of higher-

resolution models, conditional GANs for controlled sample generation, and longer or more carefully tuned training procedures. Advanced architectures, regularisation techniques, improved evaluation strategies and additional enhancements could further increase realism and diversity in the generated samples.

References

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